

# GROCERYsim ABM

## User Manual & Reference Guide

Version 2.1 | May 2026

*Agent-Based Model for Consumer Behaviour & Supply Chain Stress-Testing*

## Table of Contents

---

|                                              |           |
|----------------------------------------------|-----------|
| <b>1. Introduction &amp; Overview</b>        | <b>3</b>  |
| <b>2. Installation &amp; Getting Started</b> | <b>4</b>  |
| <b>3. Application Layout at a Glance</b>     | <b>5</b>  |
| <b>4. Sidebar: Every Control Explained</b>   | <b>6</b>  |
| <b>5. Tab 1 — Data &amp; Population</b>      | <b>9</b>  |
| <b>6. Tab 2 — Interactive Demo</b>           | <b>11</b> |
| <b>7. Tab 3 — Scientific Analysis</b>        | <b>13</b> |
| <b>8. Tab 4 — Food Waste</b>                 | <b>15</b> |
| <b>9. Tab 5 — Per-Product Deep Dive</b>      | <b>16</b> |
| <b>10. Tab 6 — Policy Analysis</b>           | <b>17</b> |
| <b>11. Tab 7 — Stakeholder View</b>          | <b>19</b> |
| <b>12. Tab 8 — Sensitivity Analysis</b>      | <b>20</b> |
| <b>13. Tab 9 — Export</b>                    | <b>21</b> |
| <b>14. How to Interpret the Charts</b>       | <b>22</b> |
| <b>15. Understanding Consumer Archetypes</b> | <b>25</b> |
| <b>16. Policy Levers: Detailed Reference</b> | <b>26</b> |
| <b>17. Model Theory &amp; Key Equations</b>  | <b>28</b> |
| <b>18. Glossary of Key Terms</b>             | <b>30</b> |
| <b>19. Troubleshooting &amp; FAQ</b>         | <b>33</b> |

# 1. Introduction & Overview

GROCERYsim ABM is a scientific simulation platform that models the behaviour of grocery consumers and the supply chain that serves them. It uses **Agent-Based Modelling (ABM)** — a technique where thousands of virtual shoppers each make their own decisions every day, producing realistic market dynamics from the bottom up.

Developed under the **SecureFood** project (Horizon Europe grant No. 101136583) at the IAMO XR Lab, it was calibrated with real DCE survey data from Finnish dairy consumers.

## 1.1 What Can GROCERYsim Do?

- Simulate daily grocery shopping for a configurable number of consumers, each with their own income, preferences, and habits.
- Run a **Baseline** (business as usual) alongside a **Crisis** scenario (price shock, supply disruption, or both).
- Model a **full crisis arc**: pre-crisis calm → active crisis → gradual recovery, with prices and supply returning to normal after a configurable duration.
- Automatically size the store to match your consumer count — a small neighbourhood shop is configured differently from a large hypermarket.
- Evaluate food policy interventions: fat taxes, domestic subsidies, supply shocks, nutritional labelling.
- Quantify food waste by product, category, and cause.
- Analyse consumer welfare across income brackets — who suffers most in a crisis?
- Run Monte Carlo simulations (multiple independent runs) to produce statistically robust results with confidence intervals.
- Conduct global sensitivity analysis to rank which parameters matter most.

## 1.2 What Is Agent-Based Modelling?

In a conventional economic model all consumers are represented as a single 'average' person. ABM instead creates thousands of individual virtual agents, each with their own characteristics. A budget-conscious student behaves differently from an environmentally motivated professional. When a crisis hits, some consumers panic-buy; others switch brands; others simply spend less. These differences produce outcomes — stockouts, price sensitivity, waste — that an average-consumer model would miss entirely.

**Tip** You do not need to understand the maths to use GROCERYsim. The simulation handles all calculations automatically. This manual explains every number in plain language.

## 1.3 Data Requirements

| Parameter / Control                           | What it means and how to use it                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Firestore Export JSON</b>                  | A JSON file exported from Firestore Realtime Database. Contains participant survey responses, DCE preference scores, demographic data (age, gender, income), and product basket data from the grocery game experiment. |
| <b>Product Catalogue master_products.json</b> | Exported from the Unity grocery game. Lists every product: name, category, fat content, organic/domestic flags, shelf life, base price, and storage capacity.                                                          |

**Note** No real data yet? Use **Demo Mode**. The app generates a synthetic population from default parameters so you can explore the simulation immediately.

## 2. Installation & Getting Started

### 2.1 Prerequisites

- **Python 3.10 or later** — download from [python.org](https://python.org)
- **pip** — included with Python; installs libraries
- A **terminal / command prompt** (Terminal on Mac, Command Prompt on Windows)
- Approximately **500 MB** of free disk space

### 2.2 Step-by-Step Setup

#### Step 1 — Open a terminal

Mac: Cmd+Space, type Terminal, Enter. Windows: Win+R, type cmd, Enter.

#### Step 2 — Navigate to the project folder

```
cd path/to/GROCERYsim
```

#### Step 3 — Install all dependencies

```
pip install -r requirements.txt  
(Takes 2-5 minutes the first time.)
```

#### Step 4 — Launch the application

```
streamlit run app.py  
Your browser opens at http://localhost:8501
```

#### Step 5 — Keep the terminal open

Do not close the terminal – it runs the server.

### 2.3 Library Reference

| Parameter / Control        | What it means and how to use it                          |
|----------------------------|----------------------------------------------------------|
| <b>streamlit</b>           | Browser-based user interface.                            |
| <b>mesa</b>                | Agent-Based Modelling framework — the simulation engine. |
| <b>pandas / numpy</b>      | Data manipulation and numerical computation.             |
| <b>plotly / matplotlib</b> | Interactive and static charts.                           |
| <b>scikit-learn</b>        | K-Means clustering for behavioural archetype assignment. |
| <b>scipy</b>               | Statistical tests (t-tests, confidence intervals).       |
| <b>SALib</b>               | Sobol global sensitivity analysis.                       |
| <b>reportlab / fpdf</b>    | PDF export for policy briefs.                            |

### 3. Application Layout at a Glance

When you open GROCERYsim you see two areas: the **Sidebar** on the left (all controls) and the **Main Area** on the right (nine result tabs).

#### 3.1 Sidebar Sections

| Parameter / Control       | What it means and how to use it                                           |
|---------------------------|---------------------------------------------------------------------------|
| <b>General</b>            | Duration, start month, consumer count, auto-calibrated store tier display |
| <b>Logistics</b>          | Reorder point, restock target, delivery lead time                         |
| <b>Crisis Scenario</b>    | Crisis start day, price inflation, supply disruption, crisis duration     |
| <b>Consumer Behaviour</b> | Panic sensitivity, hoarding factor                                        |
| <b>Monte Carlo</b>        | Number of independent simulation replications                             |
| <b>Policy Scenarios</b>   | Fat tax, domestic subsidy, supply shock, nutritional labelling            |

#### 3.2 The Nine Tabs

| Parameter / Control              | What it means and how to use it                                                |
|----------------------------------|--------------------------------------------------------------------------------|
| <b>1 — Data &amp; Population</b> | Upload files and build the consumer population. Always start here.             |
| <b>2 — Interactive Demo</b>      | Live animation: Baseline vs Crisis revenue, supply chain, behavioural drift.   |
| <b>3 — Scientific Analysis</b>   | Monte Carlo workflow with AI inventory optimisation and confidence intervals.  |
| <b>4 — Food Waste</b>            | Waste breakdown by product, category, reason, and scenario.                    |
| <b>5 — Per-Product</b>           | Detailed stock, revenue, and waste charts for individual products.             |
| <b>6 — Policy Analysis</b>       | Compare policy interventions to baseline. Download formatted PDF brief.        |
| <b>7 — Stakeholder View</b>      | Executive dashboard with traffic-light indicators and auto-generated findings. |
| <b>8 — Sensitivity Analysis</b>  | Sobol index ranking of which parameters influence outcomes most.               |
| <b>9 — Export</b>                | Download all simulation data as CSV files.                                     |

## 4. Sidebar: Every Control Explained

Every sidebar control is explained below in plain language, including what happens in the model when you change it.

### 4.1 General Parameters

| Parameter / Control                              | What it means and how to use it                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Duration (Days)</b>                           | Total simulated days. Range: 7-1,825 (5 years). Default: 60. Short runs (30-60 days) are ideal for exploring crisis effects. Longer runs (180+ days) reveal seasonal patterns and slow behavioural drift.                                                                                                                                                    |
| <b>Start Month</b>                               | Calendar month (1-12) the simulation begins. Month 12 triggers winter demand peaks; Month 7 gives a summer baseline.                                                                                                                                                                                                                                         |
| <b>Base Daily Consumers</b>                      | Approximate shoppers per day. Range: 10-5,000. Default: 100. Actual daily count varies +/-30% by weekday and season. Fridays and Saturdays are busiest; Sundays quietest. This is the midweek average. <b>Changing this value automatically resizes the entire store.</b> Shelf capacity, storage capacity, and initial stock are all recalculated to match. |
| <b>Auto-calibrated Store Tier (display only)</b> | Shown immediately below the consumer input. Updates instantly: Small (under 200/day) — compact neighbourhood shop Medium (200-999/day) — mid-size supermarket Large (1,000+/day) — large-format hypermarket All capacities and initial stock are set automatically. No manual configuration needed.                                                          |

### 4.2 Logistics Parameters

These three controls govern how and when the store reorders stock.

| Parameter / Control       | What it means and how to use it                                                                                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reorder Point (%)</b>  | When storage stock falls below this percentage of maximum storage, a delivery is triggered. Range: 10-90%. Default: 30%. Higher = order earlier, more buffer stock (less stockout risk, more waste risk). Lower = order later (less waste, more stockout risk). |
| <b>Restock Target (%)</b> | The fill level the system targets when ordering. Range: 50-100%. Default: 90%. If storage is at 20% and target is 90%, the system orders enough to fill 70% of capacity. Setting too high can cause oversupply of perishable products.                          |
| <b>Lead Time (Days)</b>   | Days between placing an order and receiving it. Range: 1-14. Default: 2. Longer lead time = more buffer stock needed. During a supply disruption, incoming deliveries are also blocked.                                                                         |

**Tip** Good starting point: Reorder Point 30%, Restock Target 90%, Lead Time 2 days. Frequent stockouts? Raise Reorder Point. High waste? Lower Restock Target.

### 4.3 Crisis Scenario Parameters

The Crisis scenario runs in parallel with the Baseline from the Crisis Start Day. Setting Crisis Duration > 0 creates a temporary shock with a full recovery arc.

| Parameter / Control     | What it means and how to use it                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Crisis Start Day</b> | The day crisis conditions begin. Both runs are identical before this day. Example: Duration = 90, Start = 30 gives 30 days of normal followed by 60 of crisis. |

| Parameter / Control             | What it means and how to use it                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Price Inflation (%)</b>      | How much prices rise when the crisis starts. Range: 0-150%. Default: 25%. 25% = moderate shock (e.g. fuel price spike). 50-100% = severe crisis. 0% = no price effect. <b>The Revenue chart uses original base prices</b> so inflation does not inflate the metric — a falling Revenue line always means fewer units sold.                                                                |
| <b>Supply Disruption (Days)</b> | Days from crisis onset during which no deliveries arrive. Range: 0-30. Default: 5. Simulates a logistics blockage — port closure, road block, factory shutdown. Deliveries resume normally after this many days.                                                                                                                                                                          |
| <b>Crisis Duration (Days)</b>   | How many days crisis conditions last before normalising. Default: 0. <b>0 = crisis runs until end of simulation.</b> Set > 0 to model a temporary shock. The simulation tracks three phases: Pre — before the Crisis Start Day Active — crisis conditions in effect Recovery — prices reverted, supply resumed, panic decaying A blue vertical line marks the Recovery Day on all charts. |

#### 4.4 Consumer Behaviour Parameters

| Parameter / Control      | What it means and how to use it                                                                                                                                                                                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Panic Sensitivity</b> | How strongly consumers react to price rises and low stock. Range: 0.0-1.0. Default: 0.5. At 0.0: consumers shop normally regardless of conditions. At 1.0: even small rises trigger maximum panic. Panic causes stockpiling and hoarding, emptying shelves. Panic decays 0.10/day in Recovery phase; 0.05/day during active crisis. |
| <b>Hoarding Factor</b>   | Multiplier on purchase quantity when a consumer is panicking. Range: 1.0-3.0. Default: 1.5. 1.5 = panicking consumers buy 50% more than usual. Separate from stockpiling — both run simultaneously.                                                                                                                                 |

#### 4.5 Monte Carlo Runs

| Parameter / Control         | What it means and how to use it                                                                                                                                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monte Carlo Runs (N)</b> | Number of independent simulation replications per scenario. Range: 3-100. Default: 10. Each run uses a different random seed. Averaging many runs produces a robust mean and a 95% confidence interval. N = 10: good for exploration. N = 20-30: publication-quality. N > 50: very precise but slow. |

## 5. Tab 1 — Data & Population

**Always visit this tab first.** It loads your files and builds the consumer population that every other tab uses.

### 5.1 Step-by-Step Instructions

#### Step 1 — Upload Firebase Export JSON

Click the left upload box. Select your Firebase JSON. This tells the model about real participants: age, gender, income, preferences, and product choices from the game.

#### Step 2 — Upload Product Catalogue JSON

Click the right upload box. Select master\_products.json. This defines every product: name, category, fat content, organic/domestic status, shelf life, price, capacity.

#### Step 3 — Set Population Pool Size

How many synthetic consumer agents to create. Range: 100-50,000. Default: 2,000. Larger pools give smoother results.

#### Step 4 — Choose Number of Behavioural Archetypes

Select 2, 3, 4, or 5 K-Means clusters. Default: 4 (the four validated archetypes: Price Champion, Green Buyer, Health Optimizer, Habitual Buyer).

#### Step 5 — Click 'Process Data and Build Population'

Parses your files, clusters participants by preference scores, assigns archetypes, and bootstraps the population. Usually takes a few seconds.

### 5.2 Reading the Population Dashboard

| Metric                     | What it measures                                                                                                         | Healthy range / target        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Summary Cards              | Four KPI cards: Real Participants, Synthetic Agents, Pool Size, Products. Verify all are non-zero before proceeding.     | All four non-zero             |
| Age Distribution           | Histogram of participant ages. Check it reflects your target population.                                                 | Matches your study population |
| Gender / Income            | Pie and bar charts. Low-income agents are most budget-constrained and welfare-vulnerable in a crisis.                    | —                             |
| Archetype Radar            | Polygon shapes per archetype on the radar. Well-separated shapes = good clustering. Similar shapes = try fewer clusters. | Clearly distinct shapes       |
| Product Catalogue          | Interactive table of all loaded products. Verify names match exactly between your Firebase file and this list.           | Every product visible         |
| Factor Score Distributions | Box plots of each preference factor. Wide spreads = diverse, realistic population.                                       | Wide spreads are good         |

**Important** If participants are skipped, their basket contained products not in the catalogue. Product names must match exactly — check capitalisation, spaces, and encoding.



## 6. Tab 2 — Interactive Demo

Runs a single paired simulation (Baseline vs Crisis) and animates the revenue chart in real time. The quickest way to see how the model behaves.

### 6.1 Controls

| Parameter / Control        | What it means and how to use it                                          |
|----------------------------|--------------------------------------------------------------------------|
| Animation Speed            | Delay per simulated day. 0.0 = instant. 0.05 = smooth flowing animation. |
| Run Comparative Simulation | Starts both runs simultaneously, day by day.                             |

### 6.2 Reading the Live Revenue Chart

The chart shows **constant-price revenue** (units sold x original base price). Inflation does not affect this metric — a falling line always means fewer units sold.

| Metric               | What it measures                                                      | Healthy range / target               |
|----------------------|-----------------------------------------------------------------------|--------------------------------------|
| Green solid line     | Baseline revenue — no crisis, no policy. Your reference.              | Roughly flat or seasonal oscillation |
| Red dashed line      | Crisis revenue. Should fall after the Crisis Start Day.               | Falls below Baseline after start     |
| Orange vertical line | Crisis Start Day.                                                     | —                                    |
| Blue vertical line   | Recovery Day — crisis conditions ended (only if Crisis Duration > 0). | —                                    |
| Gap between lines    | The revenue cost of the crisis. Larger gap = larger impact.           | Smaller gap = more resilient system  |

### 6.3 Diagnosing Revenue Gap Patterns

| Parameter / Control                     | What it means and how to use it                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Sudden large drop at crisis start       | Supply disruption emptied shelves fast. Reduce Supply Disruption Days or increase Reorder Point.                                                |
| Gap widens over time                    | Panic and hoarding compounding. Reduce Panic Sensitivity or Hoarding Factor.                                                                    |
| Crisis line recovers after Recovery Day | Expected and healthy — panic decaying, supply normalised. Faster recovery = more resilient.                                                     |
| Crisis line above Baseline              | Check you are viewing 'Revenue' (constant-price), not 'NominalRevenue'. Nominal revenue can rise with inflation even when fewer units are sold. |

### 6.4 Post-Simulation Charts

| Parameter / Control   | What it means and how to use it                                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply Chain Timeline | Per-product panel. Shelf stock, storage stock, order pipeline. Orange dashed = shelf refill threshold. Red X = expired waste. Yellow diamond = near-expiry discount. Dotted line = pipeline stock. |
| Delivery Log          | Every supply delivery: date, product, ordered qty, received qty.                                                                                                                                   |
| Revenue Sankey        | Total revenue split by product category. Wider flow = more revenue.                                                                                                                                |

| Parameter / Control      | What it means and how to use it                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Behavioural Drift</b> | Consumer preference weights over time. Flat = stable. Rising/falling lines = learning from purchase history (visible after 60+ days). |

## 7. Tab 3 — Scientific Analysis

The core research workflow. Runs multiple independent replications (Monte Carlo) to produce robust results with confidence intervals, plus AI inventory optimisation.

**Tip** Set Monte Carlo Runs (N) in the sidebar first. N = 10 for exploration; N = 30 for publication-quality results.

### 7.1 The Four-Step Workflow

#### Step 1 — Establish Baseline

Runs N independent baseline simulations. Produces a mean revenue curve with a 95% confidence band, and AI inventory recommendations based on waste and stockout patterns.

#### Step 2 — Review AI Inventory Suggestions

Compares current storage capacities against AI-recommended values. High waste product = smaller capacity recommended. Frequent stockouts = larger capacity recommended. Accept or skip.

#### Step 3 — Baseline Comparison

If you accepted AI recommendations, re-runs the baseline with optimised capacities and shows a before/after comparison.

#### Step 4 — Crisis Simulation and Impact Analysis

Runs N crisis simulations. Results compared to baseline: revenue impact, waste change, consumer welfare metrics, with statistical confidence of differences.

### 7.2 Key Statistical Outputs

| Metric                     | What it measures                                                                                                  | Healthy range / target                |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>95% CI Band</b>         | Shaded area around mean revenue. Range where the true mean falls 95% of the time across many simulations.         | Narrower = more consistent            |
| <b>Revenue Loss %</b>      | Percentage reduction in mean daily revenue during the crisis period, compared to the same period in the baseline. | 0% = no impact; more negative = worse |
| <b>Lost Sales</b>          | Units consumers tried to buy but could not (stockout).                                                            | Near 0 in baseline                    |
| <b>Fulfillment Rate</b>    | Percentage of purchase requests successfully completed.                                                           | Above 95%                             |
| <b>Budget Exhaustion %</b> | Share of shoppers who spent their entire daily budget.                                                            | Below 10% baseline                    |
| <b>Food Stressed %</b>     | Share of low-income consumers who could not meet nutritional minimum.                                             | Near 0%                               |

### 7.3 Reading Confidence Bands

- **Narrow bands** — consistent results. Typical with low panic sensitivity and stable demand.
- **Wide bands** — high variability. Increase N, or check if Panic Sensitivity is very high.
- **Overlapping Baseline and Crisis bands** — the crisis effect may not be statistically significant. Try stronger inflation or longer disruption.

- **Non-overlapping bands after Crisis Start Day** — clear, statistically significant effect. The gap between band centres is your best estimate of true impact.

## 8. Tab 4 — Food Waste

Shows all waste events from the most recent simulation. Waste occurs as: **Expiry** (product reaches shelf-life limit) or **Discard** (near-expiry units removed before they fully expire).

### 8.1 Charts Available

| Parameter / Control      | What it means and how to use it                                                                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste by Scenario</b> | Baseline vs Crisis total waste. A crisis can reduce waste (shelves empty from stockouts) or increase it (consumers stop buying expensive products). |
| <b>Waste by Product</b>  | Which products contribute most. Short shelf-life, slow-demand products usually dominate.                                                            |
| <b>Waste by Category</b> | Aggregated by category. Identifies which category needs a capacity adjustment.                                                                      |
| <b>Waste by Reason</b>   | Expired vs Discarded. High Expired waste = over-ordering or Restock Target too high.                                                                |
| <b>Daily Waste Trend</b> | Waste per day. Spikes typically follow large deliveries.                                                                                            |

### 8.2 Reducing Waste — Practical Tips

- High **Expiry** waste: over-ordering. Reduce Restock Target, or use AI suggestions in Tab 3.
- Waste **spikes after deliveries**: Restock Target too high — store fills beyond what it can sell.
- High waste + low sales for a product: reduce initial stock or remove it from the catalogue.
- Short shelf-life products (fresh milk, cheese) need lower storage targets than dry goods.

## 9. Tab 5 — Per-Product Deep Dive

Drill into the detailed performance of any individual product. Use the dropdown to select it.

| Parameter / Control      | What it means and how to use it                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stock Trajectory</b>  | Shelf stock and storage stock over time for both scenarios. Stock = zero means stockout. Large oscillations = uneven ordering.     |
| <b>Daily Revenue</b>     | Revenue from this product per day. Sharp drop in Crisis = consumers switched away due to price rise.                               |
| <b>Daily Waste</b>       | Units wasted per day. Regular spikes often match the delivery schedule.                                                            |
| <b>Daily Lost Sales</b>  | Units that could not be sold (empty shelves). Persistent non-zero values mean a higher Reorder Point or lower Lead Time is needed. |
| <b>Summary KPI Cards</b> | Total revenue, waste, lost sales, mean fulfillment — both scenarios.                                                               |

### 9.1 Diagnosing Problems

| Metric                                   | What it measures                                  | Healthy range / target                |
|------------------------------------------|---------------------------------------------------|---------------------------------------|
| <b>Stock hits zero regularly</b>         | Reorder Point too low, or Lead Time too long      | Raise Reorder Point; reduce Lead Time |
| <b>Large waste spikes after delivery</b> | Restock Target too high for this product's demand | Reduce Restock Target                 |
| <b>Lost sales spike at crisis start</b>  | Supply disruption emptied buffer stock            | Expected; increase buffer stock       |
| <b>Revenue drops more than waste</b>     | Consumers switching to cheaper products           | Expected with high inflation          |

## 10. Tab 6 — Policy Analysis

Compares a Baseline (no intervention) against a Policy scenario with one or more levers applied. The core tool for evaluating interventions.

### 10.1 Workflow

#### 1. Configure policy levers

In the sidebar, expand Policy Scenarios. Enable checkboxes for Fat Tax, Subsidy, Supply Shock, or Nutritional Labelling. Adjust slider values.

#### 2. Click 'Run Policy Analysis'

Both a baseline and a policy run execute.

#### 3. Review results

Charts and a KPI comparison table appear. An auto-generated narrative summarises key changes.

#### 4. Download Policy Brief PDF

Exports a formatted report with narrative, KPI table, charts, and income breakdown.

#### 5. Save and compare scenarios

Click 'Save This Scenario'. Change settings and run again. Up to 10 saved scenarios can be compared simultaneously.

### 10.2 Policy KPI Metrics Explained

| Metric               | What it measures                                                                                     | Healthy range / target |
|----------------------|------------------------------------------------------------------------------------------------------|------------------------|
| Revenue/day (EUR)    | Mean daily store revenue.                                                                            | Stable or increasing   |
| Waste/day (units)    | Mean daily units wasted.                                                                             | As low as possible     |
| CO2 Total/day (kg)   | Daily CO2 from sales and waste. Domestic organic = 0.8 kg/unit; imported conventional = 2.2 kg/unit. | Lower is better        |
| Import Dep. %        | Percentage of sales from imported products.                                                          | Lower = more resilient |
| Budget Exhaustion %  | Share of shoppers who spent their entire budget. A fat tax raises this for low-income groups.        | Below 10% baseline     |
| Food Stressed %      | Low-income consumers who could not meet nutritional minimum.                                         | Near 0%                |
| Fulfillment %        | Percentage of purchase requests completed.                                                           | Above 95%              |
| Mean Fat Purchased % | Average fat content of purchases. A fat tax should reduce this.                                      | Depends on goal        |

### 10.3 Reading Income Bracket Charts

Every chart is broken down by **Low / Mid / High** income. Key patterns:

- **Fat tax:** budget exhaustion rises most for Low-income. A much taller red bar = regressive policy.
- **Domestic subsidy:** fulfillment rises fairly uniformly — affordable domestic products benefit all groups.
- **Supply shock:** food stress spikes most severely for Low-income consumers with least budget flexibility.

- **Nutritional labelling:** mean fat falls gradually; strongest for Health Optimizers. Allow 90+ days.



## 11. Tab 7 — Stakeholder View

Executive dashboard for non-technical audiences — policymakers, retail managers, or funding bodies. Presents the most important findings without complex charts.

| Parameter / Control        | What it means and how to use it                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Executive KPI Cards</b> | Six numbers: Baseline daily revenue, crisis-period revenue, percentage change, total food waste, lost sales, and recovery speed.            |
| <b>Revenue Overview</b>    | Simplified Baseline vs Crisis line chart — clean for presentations.                                                                         |
| <b>Supply Chain Health</b> | Traffic-light per product: Green = adequate, Amber = low, Red = critical/stockout.                                                          |
| <b>Key Findings</b>        | Auto-generated bullets: largest revenue drop, most affected product, most vulnerable income group, crisis phase summary.                    |
| <b>Recommendations</b>     | Actionable suggestions from simulation outcomes, e.g. 'Increase Reorder Point for product X' or 'Crisis hits low-income consumers hardest.' |

**Note** Requires a completed simulation from Tab 2 or Tab 3. Run at least one simulation first.

## 12. Tab 8 — Sensitivity Analysis

Answers the question: *Which parameters matter most?* Uses Sobol variance-based global sensitivity analysis to rank each parameter by its influence on a chosen output metric.

### 12.1 Understanding Sobol Indices

| Metric                  | What it measures                                                                                | Healthy range / target                          |
|-------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>S1 (First-order)</b> | Fraction of output variance explained by this parameter acting alone, with others at average.   | High S1 = independently important               |
| <b>ST (Total-order)</b> | Fraction of output variance explained by this parameter including all interactions with others. | ST > S1 = important mainly through interactions |
| <b>ST - S1 gap</b>      | How much of the parameter's effect comes from interactions. Large gap = strong interactions.    | —                                               |

**Example:** Panic Sensitivity with S1 = 0.05 but ST = 0.35 means it barely affects output alone, but through interactions with Hoarding Factor and Supply Disruption it has a large combined effect.

### 12.2 Using the Tab

| Parameter / Control             | What it means and how to use it                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Select Output Metric</b>     | Choose: Revenue, Waste, Lost Sales, Fulfillment Rate, Budget Exhaustion, or CO2.                   |
| <b>Sample Size (N)</b>          | Model evaluations for the Sobol analysis. Recommended: 64-256 for exploration; 512+ for precision. |
| <b>Run Sensitivity Analysis</b> | Executes the calculation. May take several minutes for large N.                                    |
| <b>Results Bar Chart</b>        | Parameters sorted by influence. S1 = darker bar; ST = lighter bar.                                 |
| <b>Download CSV</b>             | Export the full index table for R, Excel, or Python.                                               |

**Tip** Run on Revenue first. If Panic Sensitivity / Hoarding Factor dominate: consumer behaviour drives the crisis. If Lead Time / Supply Disruption dominate: the supply chain is the bottleneck.

## 13. Tab 9 — Export

CSV download buttons for all simulation datasets, ready for Excel, R, or Python.

| Parameter / Control             | What it means and how to use it                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Simulation Results CSV</b>   | One row per day per scenario. Key columns: Day, Scenario, Revenue (constant-price), NominalRevenue, Waste, LostSales, Sales, Consumers, PanicLevel, AvgPrice, <b>CrisisPhase</b> , ScenarioEndDay, <b>StoreTier</b> , CO2Total, ImportDepPct, BudgetExhaustionRate, FoodStressedPct, FulfillmentRate, MeanFatPurchased, DailyLossStockout, DailyLossPrice, plus income-bracket welfare columns. |
| <b>Stock Data CSV</b>           | One row per product per day: shelf stock, storage stock, pipeline quantity.                                                                                                                                                                                                                                                                                                                     |
| <b>Supply Chain Log CSV</b>     | All delivery events: date, product, ordered qty, received qty.                                                                                                                                                                                                                                                                                                                                  |
| <b>Food Waste Log CSV</b>       | All waste events: day, product, category, reason (expired/discarded), units.                                                                                                                                                                                                                                                                                                                    |
| <b>Policy Comparison CSV</b>    | Available from Tab 6. One row per day for Baseline and Policy.                                                                                                                                                                                                                                                                                                                                  |
| <b>Sensitivity Analysis CSV</b> | Available from Tab 8. S1 and ST Sobol indices per parameter.                                                                                                                                                                                                                                                                                                                                    |

**Note** The **CrisisPhase** column contains 'pre', 'active', or 'recovery'. Use it to filter and analyse each phase of the crisis separately in your own statistical software.

## 14. How to Interpret the Charts

### 14.1 Revenue Chart

| Parameter / Control              | What it means and how to use it                                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Green solid line</b>          | Baseline constant-price revenue. Roughly flat or seasonal is normal.                                                                                              |
| <b>Red dashed line</b>           | Crisis revenue. Should fall after the Crisis Start Day.                                                                                                           |
| <b>Orange vertical line</b>      | Crisis Start Day.                                                                                                                                                 |
| <b>Blue vertical line</b>        | Recovery Day — crisis conditions ended (only if Crisis Duration > 0).                                                                                             |
| <b>Shaded confidence band</b>    | 95% CI across N Monte Carlo runs. Overlap = no significant difference.                                                                                            |
| <b>Revenue vs NominalRevenue</b> | Always compare using 'Revenue' (constant-price). 'NominalRevenue' rises with inflation even when fewer units are sold — misleading for cross-scenario comparison. |

### 14.2 Supply Chain Stock Chart

| Parameter / Control           | What it means and how to use it                                   |
|-------------------------------|-------------------------------------------------------------------|
| <b>Blue area</b>              | Shelf stock — what consumers can currently buy.                   |
| <b>Green area</b>             | Storage stock — back-of-store reserve.                            |
| <b>Dotted line</b>            | Pipeline stock — ordered, not yet delivered.                      |
| <b>Orange dashed line</b>     | Shelf refill threshold — shelves restocked when below this.       |
| <b>Red dashed line</b>        | Storage reorder point — new order placed when storage falls here. |
| <b>Red X markers</b>          | Expired waste events.                                             |
| <b>Yellow diamond markers</b> | Near-expiry discount events — units price-reduced to sell faster. |

### 14.3 Income Bracket Charts (Policy Analysis)

| Parameter / Control        | What it means and how to use it                                       |
|----------------------------|-----------------------------------------------------------------------|
| <b>Blue bars / lines</b>   | High-income consumers. Usually least affected by price shocks.        |
| <b>Orange bars / lines</b> | Mid-income consumers. Moderate policy impact.                         |
| <b>Red bars / lines</b>    | Low-income consumers. Most vulnerable. Watch these for equity impact. |

### 14.4 Consumer Archetype Radar Chart

- Polygon near outer edge on **Price** axis = highly price-sensitive consumers.
- Polygon near outer edge on **Organic** or **Domestic** = strong environmental preference.
- Very similar polygon shapes across archetypes = try fewer K-Means clusters.
- Area of each polygon = overall preference intensity of that archetype.

### 14.5 Sensitivity Analysis Bar Chart

- Tall S1 bar = this parameter independently explains a large share of output variance.
- Tall ST bar with small S1 = important mainly through interactions with other parameters.

- Short S1 and ST bars = little effect on the chosen output metric.
- Sum of all S1 indices is always below 1. Sum of all ST indices can exceed 1 (interactions counted multiple times).

## 15. Understanding Consumer Archetypes

GROCERYsim uses K-Means clustering on DCE preference scores to group consumers into four behavioural archetypes. Each agent retains its archetype-based preferences throughout the simulation, with gradual daily learning drift.

| Archetype               | Key traits                                                              | Under crisis                                                       | Under policy                                                      |
|-------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Price Champion</b>   | Maximises value; highly price-sensitive; typically low-to-mid income.   | Most likely to panic-buy. Switches to cheapest available products. | Most hurt by fat tax; benefits most from domestic subsidy.        |
| <b>Green Buyer</b>      | Strong preference for organic and domestic; moderate price sensitivity. | Switches away from imports first; tolerates moderate price rises.  | Amplifies CO2 reduction and import drop under subsidy.            |
| <b>Health Optimizer</b> | Prioritises low fat and nutritional quality; less price-sensitive.      | May switch to lower-fat alternatives; less likely to panic-buy.    | Most responsive to nutritional labelling.                         |
| <b>Habitual Buyer</b>   | Brand and habit loyal; low price and policy sensitivity; slow drift.    | Absorbs mild shocks. Responds only to extreme pressure.            | Resistant to short-term interventions. Needs 180+ days for drift. |

### 15.1 Behavioural Learning

Each agent updates preference weights daily (rate  $\sim 0.015/\text{day}$ ). Satisfying purchases reinforce preferences; failed purchases (stockout, price shock) weaken them. Home inventory is tracked persistently — consumers reduce next-day purchases when already stocked up, preventing unlimited hoarding.

## 16. Policy Levers: Detailed Reference

### 16.1 Fat Tax

Adds a price surcharge to products with fat content at or above a threshold.

| Parameter / Control | What it means and how to use it                                           |
|---------------------|---------------------------------------------------------------------------|
| Enable Fat Tax      | Check to activate.                                                        |
| Fat % Threshold     | Products at or above this fat% are taxed. Range: 0.5-5.0%. Default: 3.5%. |
| Surcharge Rate (%)  | Percentage added to base price. Range: 5-50%. Default: 20%.               |

**Expected effects:** Revenue falls initially. Over 60-90 days, preference learning shifts demand toward lower-fat products. Strongest for Price Champions; weakest for Habitual Buyers.

### 16.2 Domestic / Organic Subsidy

Applies a price discount to domestic and/or organic products.

| Parameter / Control | What it means and how to use it                                   |
|---------------------|-------------------------------------------------------------------|
| Enable Subsidy      | Check to activate.                                                |
| Subsidy Target      | Domestic only, organic only, or both.                             |
| Discount Rate (%)   | Price reduction on eligible products. Range: 5-40%. Default: 15%. |

**Expected effects:** Import dependency and CO2 footprint decrease. Welfare improvement is fairly uniform across income brackets.

### 16.3 Domestic Supply Shock

Blocks deliveries of domestic products for a configurable period.

| Parameter / Control   | What it means and how to use it                                                     |
|-----------------------|-------------------------------------------------------------------------------------|
| Enable Supply Shock   | Check to activate.                                                                  |
| Shock Start Day       | Day the shock begins.                                                               |
| Shock Duration (Days) | Duration of the blockage. Range: 1-120. Default: 30.                                |
| Shock Severity        | Fraction of domestic deliveries blocked. 1.0 = complete; 0.5 = half. Default: 0.70. |

**Expected effects:** Import dependency and CO2 rise sharply. Low-income consumers who relied on affordable domestic products suffer most.

### 16.4 Nutritional Labelling

Simulates mandatory clearer health labelling (e.g. traffic-light labels). Implemented as a boost to health and organic preference weights.

| Parameter / Control | What it means and how to use it                                   |
|---------------------|-------------------------------------------------------------------|
| Enable Labelling    | Check to activate.                                                |
| Labelling Start Day | Day the policy takes effect.                                      |
| Health Pref. Boost  | Added to health preference weight. Range: 0.0-0.4. Default: 0.15. |

| Parameter / Control        | What it means and how to use it                                    |
|----------------------------|--------------------------------------------------------------------|
| <b>Organic Pref. Boost</b> | Added to organic preference weight. Range: 0.0-0.3. Default: 0.10. |

**Expected effects:** Mean fat purchased falls gradually. Allow 60-90+ days. Health Optimizers respond most; Habitual Buyers least.



## 17. Model Theory & Key Equations

For readers who want to understand the mechanics. Non-technical readers can skip to Section 18.

### 17.1 Consumer Utility Function

Each consumer evaluates all available products and buys the one with the highest utility score:

$$U(a,p) = w\_origin \cdot origin(p) + w\_organic \cdot organic(p) + w\_fat \cdot fat\_match(a,p) - w\_price \cdot price\_disutility(a,p)$$

| Parameter / Control     | What it means and how to use it                                                |
|-------------------------|--------------------------------------------------------------------------------|
| <b>origin(p)</b>        | 1.0 if domestic; 0.0 if imported.                                              |
| <b>organic(p)</b>       | 1.0 if organic; 0.0 otherwise.                                                 |
| <b>fat_match(a,p)</b>   | Positive if product fat matches agent's preference; negative if mismatched.    |
| <b>price_disutility</b> | Effective price divided by agent's income-normalised budget.                   |
| <b>w_* weights</b>      | Agent-specific preference weights. Initialised from DCE scores; updated daily. |

**Prospect Theory (Kahneman & Tversky 1979):** the price disutility function uses loss aversion — consumers perceive a price above their reference as a loss, weighted more heavily than an equivalent gain. The reference price adapts slowly via an exponential moving average.

### 17.2 Daily Simulation Sequence

#### 1. Seasonal/weekday scaling

Consumer count = base\_consumers x weekday\_weight x seasonal\_weight. Weekday weights: Mon 0.80, Tue 0.90, Wed 0.90, Thu 1.00, Fri 1.10, Sat 1.30, Sun 0.70.

#### 2. Shelf replenishment

If shelf stock falls below 50% of shelf capacity, units moved from storage to shelf.

#### 3. Panic update

Global panic updated from price inflation, stock levels, and media signals. Decays 0.10/day in Recovery phase; 0.05/day during active crisis.

#### 4. Reorder check

For each product: if (storage + pending deliveries) < Reorder Point threshold, an order is placed targeting the Restock Target level.

#### 5. Consumer shopping

Each active agent selects a product using the utility function, modulated by home inventory state and panic level. Panicking consumers may buy extra (hoarding).

#### 6. Expiry removal

Products exceeding shelf life are logged as food waste.

#### 7. Delivery arrival

Orders placed Lead Time days ago arrive into storage (unless supply disruption active).

## 8. Metrics recording

All KPIs saved: revenue, waste, lost sales, panic level, crisis phase, welfare metrics, etc.

### 17.3 Store Auto-Calibration

Shelf and storage capacities are computed automatically from the consumer count:

```
shelf_cover = 1.5 days (perishable) | 2.5 days (medium) | 4.0 days (dry goods)
daily_demand = base_consumers x avg_basket_qty_for_product
max_shelf = daily_demand x shelf_cover
max_storage = daily_demand x (lead_time + 4)
initial_shelf = max_shelf x 0.75
initial_storage = max_storage x 0.60
```

This ensures a 200-consumer shop is never configured the same as a 2,000-consumer hypermarket. The result is displayed as the Store Tier badge in the sidebar.

### 17.4 CO2 Footprint Factors

| Metric                       | What it measures           | Healthy range / target |
|------------------------------|----------------------------|------------------------|
| <b>Finnish Organic</b>       | 0.8 kg CO2e per unit sold. | Lowest footprint       |
| <b>Finnish Conventional</b>  | 1.2 kg CO2e per unit sold. | Low footprint          |
| <b>Imported Organic</b>      | 1.5 kg CO2e per unit sold. | Moderate footprint     |
| <b>Imported Conventional</b> | 2.2 kg CO2e per unit sold. | Highest footprint      |

## 18. Glossary of Key Terms

---

### **ABM (Agent-Based Model)**

A simulation where each individual agent follows its own rules. Collective outcomes emerge from the bottom up.

### **Active Crisis Phase**

Period between the Crisis Start Day and Recovery Day. Price inflation and supply disruption are in effect.

### **Archetype**

A cluster of consumers with similar preference profiles: Price Champion, Green Buyer, Health Optimizer, or Habitual Buyer.

### **Baseline**

The simulation with no crisis and no policy. All scenarios are compared to this.

### **Budget Exhaustion Rate**

Share of daily shoppers who spent their entire available budget.

### **CO2 Footprint**

Total CO2-equivalent from products sold and wasted. Range: 0.8-2.2 kg per unit.

### **Constant-Price Revenue**

Revenue = units sold x original base price. Unaffected by inflation — a fall always means fewer units sold.

### **Crisis Duration**

How many days crisis conditions last before prices revert and supply resumes. 0 = runs to end of simulation.

### **Crisis Phase**

One of three states: 'pre' (before crisis), 'active' (crisis in effect), 'recovery' (normalised, panic decaying).

### **DCE (Discrete Choice Experiment)**

Survey where respondents choose between products with varying attributes. Used to calibrate consumer preference weights.

### **Fulfillment Rate**

Percentage of consumer purchase requests successfully completed. Target: above 95%.

### **Hoarding Factor**

Multiplier on purchase quantity for panicking consumers. 1.5 = 50% extra.

### **Home Inventory**

Stock the consumer already has at home. Tracked persistently — reduces next-day purchases when high.

### **Import Dependency %**

Fraction of daily sales from imported products. Higher = more exposed to global supply disruptions.

### **Lead Time**

Days between placing a supply order and receiving the delivery.

### **Lost Sales**

Units consumers tried to buy but could not because shelves were empty.

### **Monte Carlo Simulation**

Multiple independent runs with different random seeds. Results averaged to produce robust estimates with confidence intervals.

**Nominal Revenue**

Revenue at current (inflated) prices. Rises with inflation even if fewer units sold. Not suitable for crisis comparisons — always use Constant-Price Revenue.

**Panic Level**

A 0.0-1.0 score for consumer fear. Triggers stockpiling and hoarding. Decays over time once conditions normalise.

**Prospect Theory**

Kahneman & Tversky (1979): losses feel larger than equivalent gains. Used in the price disutility function.

**Recovery Phase**

Days after Crisis Duration ends. Prices reverted, supply resumed, panic decaying at 0.10/day.

**Reorder Point**

Storage fill level (% of maximum) at which a new supply order is triggered.

**Restock Target**

Storage fill level the system aims for when placing a reorder.

**S1 / ST (Sobol Indices)**

S1 = first-order sensitivity (independent effect). ST = total-order (including interactions). Used in Sensitivity Analysis.

**Shelf Life**

Days a product can remain on the shelf before it expires and is discarded.

**Stockout**

When shelf stock reaches zero and consumers cannot buy the product.

**Store Tier**

Auto-assigned label based on consumer count: Small (<200), Medium (200-999), Large (1,000+). Determines calibrated shelf and storage capacities.

**Supply Disruption**

A configurable period at crisis onset during which no deliveries arrive.

**Utility**

Numerical score of how desirable a product is for a given consumer. Consumers choose the highest-utility in-stock product.

## 19. Troubleshooting & FAQ

---

### **Q: The app does not start / 'streamlit: command not found'**

A: Run: `pip install -r requirements.txt`. If multiple Python installs: `python -m streamlit run app.py`

### **Q: Processing failed: 'No products matched'**

A: Product names in your Firebase data do not match `master_products.json`. Check capitalisation, trailing spaces, and encoding — names must be identical.

### **Q: Participants were skipped**

A: Normal if a small number had no matching basket entries. If over 50% are skipped, your data files are misaligned — check product name consistency.

### **Q: Simulation is very slow**

A: Reduce Base Daily Consumers, Duration, or Monte Carlo Runs (N). Quick test: 30 days, 50 consumers, N = 5.

### **Q: Revenue drops to zero**

A: All products are out of stock. Set Restock Target to 95%, reduce Lead Time to 1 day, or reduce Supply Disruption Days.

### **Q: Crisis revenue is going UP instead of down**

A: Ensure you are viewing the 'Revenue' chart (constant-price), not 'NominalRevenue'. Nominal revenue rises with inflation even when fewer units are sold.

### **Q: Policy Analysis shows no difference from Baseline**

A: Ensure at least one policy lever checkbox is ticked, and that values are non-trivial (fat tax rate > 0%, shock duration > 0 days, discount > 0%).

### **Q: Recovery phase does not appear on charts**

A: Set Crisis Duration > 0 in the Crisis Scenario sidebar section. The default (0) means the crisis runs until the end of the simulation.

### **Q: Confidence bands are very wide**

A: Increase N to 20-30. Wide bands with high N = high model variability, likely from high Panic Sensitivity. Try reducing it.

### **Q: Waste is very high even in the Baseline**

A: Restock Target is too high for the demand level. Reduce it to 70-80%, or use the AI inventory suggestions in Tab 3.

### **Q: What does the 95% CI band mean?**

A: Across N runs, each day has N revenue values. The band covers the 5th to 95th percentile. If Baseline and Crisis bands overlap, the difference is not statistically significant with your current settings.

**Q: Can I use my own consumer data?**

A: Yes. Replace the Firebase export with your own participant data following the same JSON structure. Update master\_products.json to match your product catalogue.

**Q: Can I deploy the app online?**

A: Push the repo to GitHub and deploy free at <https://streamlit.io/cloud>.

---

**Credits & Acknowledgements**

GROCERYsim ABM v2.1 was developed as part of the **SecureFood** project (Horizon Europe grant No. 101136583). UI: **Streamlit**. ABM engine: **Mesa**. Sensitivity: **SALib**. PDF: **ReportLab**. Theory: Kahneman & Tversky (1979), Ajzen (1991), O'Donoghue & Rabin (1999), McCombs & Shaw (1972), FAO FIES (2016).

Document generated: May 2026. Contact: IAMO XR Lab.